Genomic Surveillance of Klebsiella pneumoniae: Antimicrobial Resistance, Virulence, and Plasmid Convergence Across Lineages

A Comparative Genomics and Phylogenetic Analysis of 21 Publicly Available Clinical Isolates

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Abstract

Klebsiella pneumoniae is a WHO critical-priority pathogen driving the global antimicrobial resistance (AMR) crisis, particularly through carbapenem-resistant strains in hospital settings. This project applied whole-genome comparative and phylogenetic methods to a curated set of 21 publicly available, human clinical K. pneumoniae genomes sourced from five independent studies (Turkiye, India, and China) to investigate the extent to which AMR determinants and virulence factors co-occur within the same lineages, and the role plasmids play in that convergence. Using Kleborate, MOB-suite, Panaroo, and IQ-TREE, the analysis identified 11 distinct sequence types (STs) among the 21 genomes. Two lineages — ST2096 and ST11 — showed genuine convergence of high virulence and high resistance scores, each carrying multiple plasmids, while ST23 and ST420 displayed the classic 'pure hypervirulence' pattern (high virulence, zero acquired resistance). Phylogenetic analysis showed that ST2096 and ST11 occupy distinct, well-separated clades on the tree, indicating that this convergence phenotype arose independently in two separate lineages rather than through clonal spread from a single ancestor — consistent with a model of plasmid-mediated convergent evolution. These findings reinforce the value of integrated genomic surveillance combining lineage typing, resistome, virulome, and mobile genetic element analysis for tracking the emergence of high-risk K. pneumoniae clones.

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1. Introduction

1.1 Background

Klebsiella pneumoniae is a Gram-negative, encapsulated, opportunistic bacterial pathogen and a leading cause of healthcare-associated infections, including pneumonia, bloodstream infections, urinary tract infections, and neonatal sepsis. It is normally a harmless gut commensal but becomes dangerous in immunocompromised patients or when introduced into sterile body sites via medical devices. The World Health Organization has designated carbapenem-resistant K. pneumoniae (CRKP) as a critical-priority pathogen, reflecting both its clinical prevalence and its growing resistance to last-resort antibiotics.

A separate and historically distinct concern is hypervirulent K. pneumoniae (hvKp), classically associated with community-acquired liver abscess even in healthy individuals, driven by capsule overproduction (rmpA/rmpA2) and enhanced iron-scavenging systems (aerobactin, yersiniabactin, salmochelin). Traditionally, hypervirulent strains were antibiotic-susceptible and resistant strains were not especially virulent. However, recent genomic surveillance literature has documented an alarming trend: the convergence of hypervirulence and carbapenem resistance within the same strains, a combination that would be exceptionally difficult to treat and highly pathogenic.

1.2 Literature Context

Review of eight recent peer-reviewed studies (2025-2026) on K. pneumoniae genomic surveillance — spanning China, Ireland, Japan, Spain, Tanzania, and India — confirmed several recurring themes relevant to this project: (1) the necessity of combining lineage typing (MLST), AMR profiling, virulence profiling, and plasmid analysis to understand transmission and risk; (2) documented cases of AMR-virulence convergence mediated by conjugative plasmids (e.g., IncFIIK2-NDM-1 plasmids); and (3) the importance of cautious interpretation — genomic similarity alone does not prove transmission without supporting epidemiological metadata. These themes directly shaped this project's design.

1.3 Why K. pneumoniae for This Project

K. pneumoniae was selected over other candidate pathogens for several reasons: it sits at the top of the WHO AMR priority list; it has rich, well-characterized genomic determinants for both resistance and virulence, allowing meaningful co-occurrence analysis; it has mature, purpose-built genomic surveillance tooling (Kleborate) actively used by real public health consortia (KlebNET-GSP); and

abundant, fully public genome data is available, avoiding the data-access barriers common to human-subject-linked datasets.

2. Research Question

Among a curated set of publicly available Klebsiella pneumoniae genomes, to what extent do antimicrobial resistance and virulence determinants co-occur within the same lineages, and what role do plasmids play in that convergence?

This question was selected because it was the theme most consistently supported across the reviewed literature, is fully answerable using public genome assemblies without requiring linked clinical/phenotype data, and maps directly onto a complete, standard genomic surveillance workflow (lineage typing, resistome, virulome, mobilome, and phylogenetics).

3. Methods

3.1 Dataset Selection

Genomes were sourced from NCBI (Assembly and BioProject databases) under the following inclusion criteria: (a) confirmed Klebsiella pneumoniae identity; (b) human-associated clinical isolates only, excluding animal/agricultural and laboratory-mutant genomes; (c) Complete Genome assembly level only, to ensure reliable gene- and plasmid-level analysis; and (d) traceable to a documented, multi-isolate BioProject with available metadata. Twenty-one genomes meeting these criteria were selected from five independent BioProjects: a carbapenem-resistant K. pneumoniae (CRKP) study from two tertiary-care hospitals in Turkiye (10 genomes), a second Turkiye clinical isolate study (4 genomes), a sequence type, resistome, and virulome study from Pune, India (3 genomes), a clinical isolate study from Peking University People's Hospital, China (2 genomes), and a study focused on the globally significant ST11 high-risk clone (2 genomes). The full accession list is provided in Appendix A.

3.2 Genome Quality Control and Annotation

Assembly quality was assessed using QUAST v5.2.0, checking genome size, GC content, N50/L50, and ambiguous base content against expected values for K. pneumoniae. All 21 genomes passed quality control. Genome annotation was performed using Prokka v1.15.6 with taxonomic parameters set to Klebsiella pneumoniae, producing standardized GFF3 annotation files required for downstream comparative genomics.

3.3 Lineage, Resistance, and Virulence Profiling

Sequence typing (multi-locus sequence typing, MLST), acquired AMR gene detection, virulence gene detection (yersiniabactin, colibactin, aerobactin, salmochelin, and rmpA/rmpA2 loci), and capsule (K/O) serotyping were performed using Kleborate v3.2.4 with the K. pneumoniae Species Complex (kpsc) preset, run directly on the assembled genome sequences.

3.4 Plasmid Analysis

Plasmid content was reconstructed and characterized from each genome assembly using MOB-suite v3.1.9 (mob_recon), which identifies and types plasmid-derived contigs and reports replicon content per genome.

3.5 Comparative (Pangenome) Analysis

Core and accessory genome content across all 21 genomes was determined using Panaroo v1.8.0 in strict clean-mode, using the Prokka-generated GFF3 files as input.

3.6 Multiple Sequence Alignment and Phylogenetics

A core genome alignment (genes present in ≥99% of isolates; 4,104 genes) was generated using Panaroo's integrated alignment function (MAFFT). A maximum-likelihood phylogenetic tree was then constructed from this alignment using IQ-TREE v3.1.3 with 1,000 ultrafast bootstrap replicates. The resulting tree was visualized using the Interactive Tree of Life (iTOL) web platform, with tips relabeled to show sequence type and original strain identifier for interpretability.

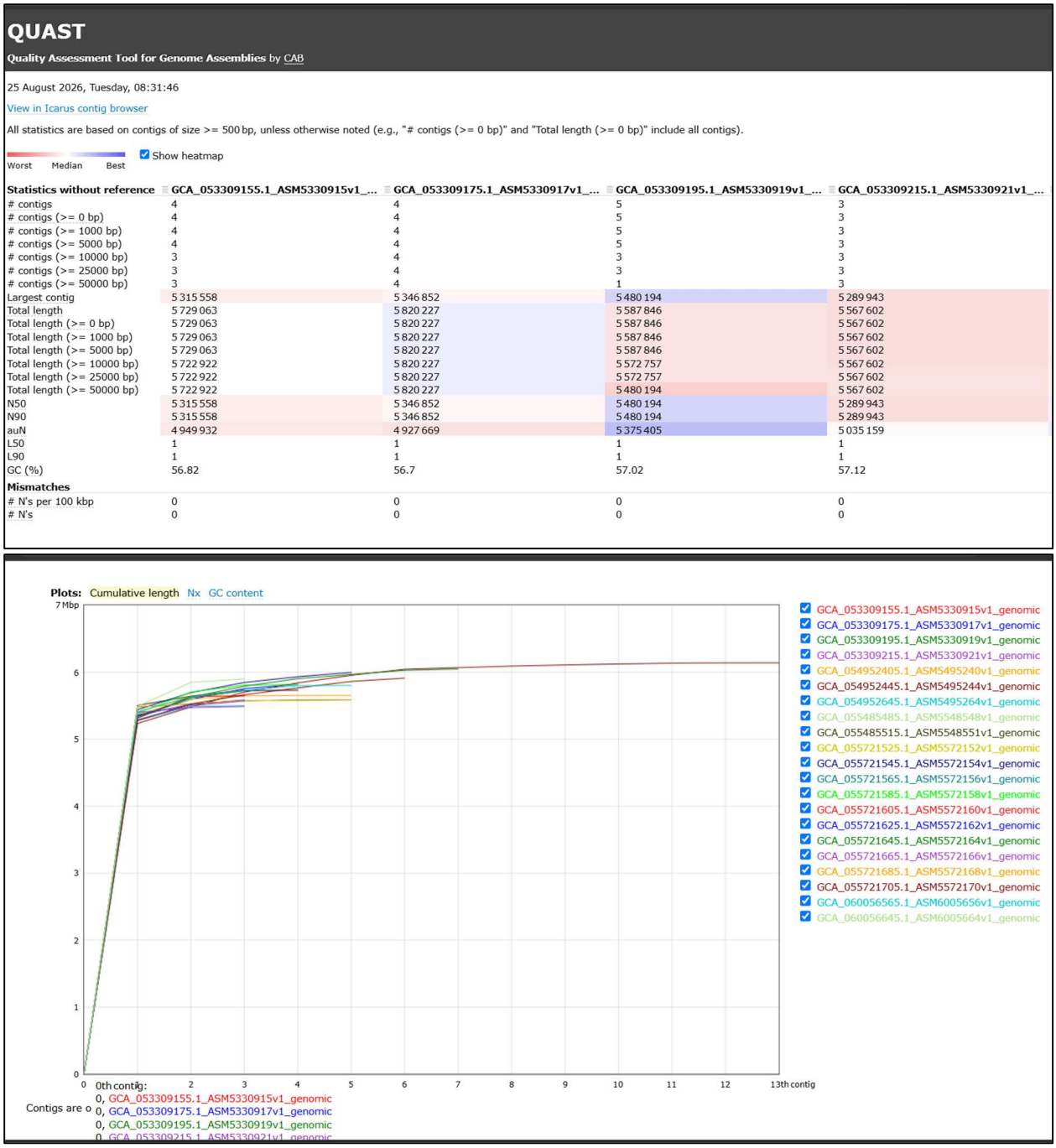
3.7 Data Integration

Results from Kleborate and MOB-suite were merged into a single master summary table using a custom Python script (pandas), enabling direct comparison of sequence type, virulence score, resistance score, and plasmid count across all 21 genomes.

4. Results

4.1 Genome Quality

All 21 genomes passed quality control: every assembly consisted of a single contiguous sequence (L50 = 1), genome sizes ranged from 5.49-6.14 Mb, and GC content ranged from 56.5-57.2%, consistent with expected values for K. pneumoniae.



QUAST

Quality Assessment Tool for Genome Assemblies by CAB

25 August 2026, Tuesday, 08:31:46

View in Icarus contig browser

All statistics are based on contigs of size >= 500 bp, unless otherwise noted (e.g., "# contigs (>= 0 bp)" and "Total length (>= 0 bp)" include all contigs).

Worst Median Best — Show heatmap

Statistics without reference	GCA_053309155.1_ASM5330915v1_...	GCA_053309175.1_ASM5330917v1_...	GCA_053309195.1_ASM5330919v1_...	GCA_053309215.1_ASM5330921v1_...
# contigs	4	4	5	3
# contigs (>= 0 bp)	4	4	5	3
# contigs (>= 1000 bp)	4	4	5	3
# contigs (>= 5000 bp)	4	4	5	3
# contigs (>= 10000 bp)	3	4	3	3
# contigs (>= 25000 bp)	3	4	3	3
# contigs (>= 50000 bp)	3	4	1	3
Largest contig	5 315 558	5 346 852	5 480 194	5 289 943
Total length	5 729 063	5 820 227	5 587 846	5 567 602
Total length (>= 0 bp)	5 729 063	5 820 227	5 587 846	5 567 602
Total length (>= 1000 bp)	5 729 063	5 820 227	5 587 846	5 567 602
Total length (>= 5000 bp)	5 729 063	5 820 227	5 587 846	5 567 602
Total length (>= 10000 bp)	5 722 922	5 820 227	5 572 757	5 567 602
Total length (>= 25000 bp)	5 722 922	5 820 227	5 572 757	5 567 602
Total length (>= 50000 bp)	5 722 922	5 820 227	5 480 194	5 567 602
N50	5 315 558	5 346 852	5 480 194	5 289 943
N90	5 315 558	5 346 852	5 480 194	5 289 943
auN	4 949 932	4 927 669	5 375 405	5 035 159
L50	1	1	1	1
L90	1	1	1	1
GC (%)	56.82	56.7	57.02	57.12
Mismatches				
# N's per 100 kbp	0	0	0	0
# N's	0	0	0	0

Plots: Cumulative length Nx GC content

7 Mbp
6
5
4
3
2
1
0
0 1 2 3 4 5 6 7 8 9 10 11 12 13th contig
0th contig:
Contigs are o 0, GCA_053309155.1_ASM5330915v1_genomic
0, GCA_053309175.1_ASM5330917v1_genomic
0, GCA_053309195.1_ASM5330919v1_genomic
0, GCA_053309215.1_ASM5330921v1_genomic

GCA_053309155.1_ASM5330915v1_genomic
GCA_053309175.1_ASM5330917v1_genomic
GCA_053309195.1_ASM5330919v1_genomic
GCA_053309215.1_ASM5330921v1_genomic
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GCA_054952645.1_ASM5495264v1_genomic
GCA_055485485.1_ASM5548548v1_genomic
GCA_055485515.1_ASM5548551v1_genomic
GCA_055721525.1_ASM5572152v1_genomic
GCA_055721545.1_ASM5572154v1_genomic
GCA_055721565.1_ASM5572156v1_genomic
GCA_055721585.1_ASM5572158v1_genomic
GCA_055721605.1_ASM5572160v1_genomic
GCA_055721625.1_ASM5572162v1_genomic
GCA_055721645.1_ASM5572164v1_genomic
GCA_055721665.1_ASM5572166v1_genomic
GCA_055721685.1_ASM5572168v1_genomic
GCA_055721705.1_ASM5572170v1_genomic
GCA_060056565.1_ASM6005656v1_genomic
GCA_060056645.1_ASM6005664v1_genomic

Figure 1. Genome assembly quality metrics across all 21 isolates (QUAST).

4.2 Genome Annotation

Prokka annotation identified between 5,100 and 5,849 predicted coding sequences (CDS) per genome, consistent with the expected gene content range for K. pneumoniae.

4.3 Sequence Types, Resistance, and Virulence

Kleborate identified 11 distinct sequence types among the 21 genomes:

Sequence Type	Genome Count	Notable Feature
ST2096	4	High virulence AND high resistance (convergence)
ST11	4	Globally significant carbapenem-resistant clone; convergence
ST307	2	Moderate resistance, low virulence
ST101	2	Zero acquired resistance despite plasmid presence
ST23	1	Highest virulence score (5), zero resistance genes
ST420	1	High virulence, zero resistance
ST15, ST985, ST16, ST14, ST147, ST1418	1 each	Single representatives

Notably, ST147 (strain KK35) showed the lowest scores across every metric measured: virulence score 0, resistance score 1, and zero detected plasmids.

4.4 Plasmid Content

MOB-suite detected plasmid content in the large majority of genomes. ST147 (KK35) was the sole exception, showing zero reconstructed plasmids, consistent with its minimal resistance/virulence profile. Genomes belonging to the convergent lineages (ST2096, ST11) each carried between 2 and 4 distinct plasmids.

4.5 Pangenome Structure

Panaroo analysis of the 21-genome pangenome identified:

Gene Category	Definition	Count
Core genes	Present in 99-100% of genomes	4,104
Soft core genes	Present in 95-99% of genomes	141
Shell genes	Present in 15-95% of genomes	2,250
Cloud genes	Present in <15% of genomes	3,188

The large cloud and shell gene categories (approximately 56% of the total ~9,683-gene pangenome) confirm K. pneumoniae's characteristically "open" pangenome, consistent with published literature, and indicate that this species' accessory genome — where AMR and virulence genes typically reside — is highly variable across strains.

4.6 Integrated Summary

Merging Kleborate and MOB-suite results into a single table revealed the following overall pattern:

ST	Virulence Score	Resistance Score	Plasmids	Pattern
ST23	5 (highest)	0	1	Pure hypervirulence
ST420	4	0	1	Pure hypervirulence
ST2096 (n=4)	3-4	2-3	2-3	True convergence
ST11 (n=4)	1-4	1-2	2-4	True convergence
ST147	0	1	0	Lowest overall
ST101 (n=2)	1	0	3	Plasmid-rich but low risk

	accession	species	ST	virulence_score	resistance_score	num_resistance_classes	num_resistance_genes	num_plasmids
2	GCA_055721685.1	Klebsiella pneumoniae	ST101	1	0	10	15	3
3	GCA_053309195.1	Klebsiella pneumoniae	ST101	1	0	10	15	3
4	GCA_060056645.1	Klebsiella pneumoniae	ST11	4	2	7	6	2
5	GCA_060056565.1	Klebsiella pneumoniae	ST11	4	2	9	10	4
6	GCA_055485515.1	Klebsiella pneumoniae	ST11	1	2	9	12	2
7	GCA_055485485.1	Klebsiella pneumoniae	ST11	1	1	9	12	2
8	GCA_055721545.1	Klebsiella pneumoniae	ST14	1	2	11	26	3
9	GCA_055721705.1	Klebsiella pneumoniae	ST1418	1	3	10	24	4
10	GCA_055721665.1	Klebsiella pneumoniae	ST147	0	1	3	3	0
11	GCA_054952445.1	Klebsiella pneumoniae	ST15	1	2	7	15	6
12	GCA_055721645.1	Klebsiella pneumoniae	ST15	1	2	9	17	6
13	GCA_055721525.1	Klebsiella pneumoniae	ST16	1	3	11	13	1
14	GCA_055721565.1	Klebsiella pneumoniae	ST2096	3	2	7	17	3
15	GCA_055721585.1	Klebsiella pneumoniae	ST2096	4	3	11	19	3
16	GCA_055721605.1	Klebsiella pneumoniae	ST2096	4	3	11	17	2
17	GCA_053309155.1	Klebsiella pneumoniae	ST2096	3	2	7	16	3
18	GCA_054952405.1	Klebsiella pneumoniae	ST23	5	0	0	0	1
19	GCA_055721625.1	Klebsiella pneumoniae	ST307	0	3	7	7	2
20	GCA_053309175.1	Klebsiella pneumoniae	ST307	0	3	7	7	3
21	GCA_054952645.1	Klebsiella pneumoniae	ST420	4	0	0	0	1
22	GCA_053309215.1	Klebsiella pneumoniae	ST985	1	2	8	12	2

Figure 2. Integrated virulence and resistance scores by sequence type.

4.7 Phylogenetic Analysis

The maximum-likelihood tree, built from the 4,104-gene core genome alignment across all 21 genomes with 1,000 ultrafast bootstrap replicates, showed that isolates sharing the same sequence type consistently formed tight, well-defined clades — confirming that MLST-based lineage assignment reflects genuine evolutionary relatedness in this dataset. Critically, the two convergent lineages, ST2096 and ST11, occupy clearly separated, non-adjacent positions on the tree.

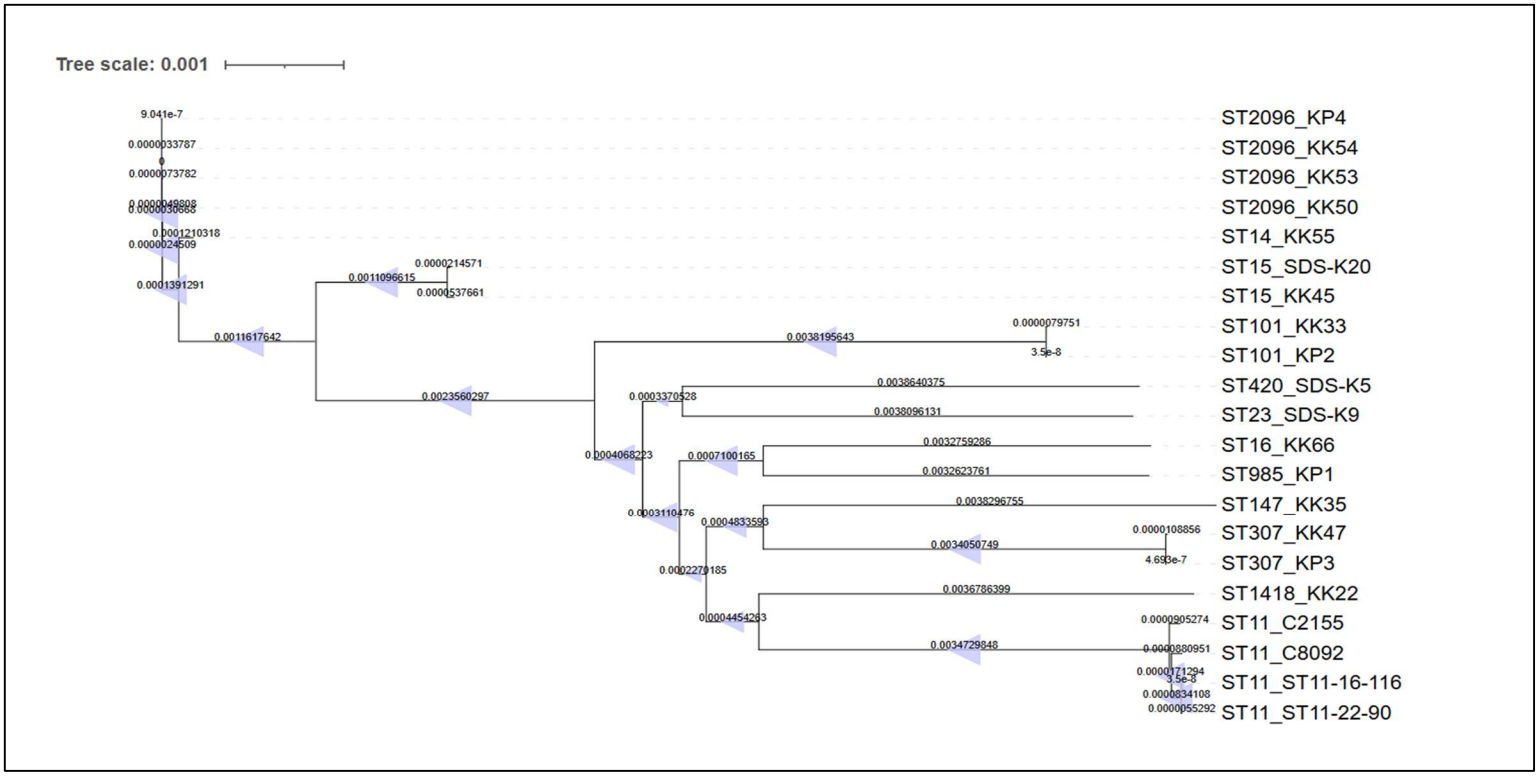


Figure 3. Maximum-likelihood phylogenetic tree of 21 K. pneumoniae genomes, core genome alignment, IQ-TREE with 1,000 ultrafast bootstrap replicates. Tips labeled as sequence type and strain identifier.

5. Discussion

This analysis provides direct, quantitative evidence addressing the project's research question. Two clear phenotype patterns emerged among the 21 genomes: a "pure hypervirulence" pattern (ST23, ST420 — high virulence score, zero acquired resistance genes) and a "convergence" pattern (ST2096, ST11 — simultaneously elevated virulence and resistance scores, each associated with multiple plasmids).

The phylogenetic placement of ST2096 and ST11 in separate, well-distinguished clades is a particularly important finding. Had these two lineages appeared adjacent or nested within one another, it would suggest the convergence phenotype arose once and spread clonally. Instead, their clear phylogenetic separation indicates that AMR-virulence convergence emerged independently at least twice within this dataset — most plausibly through independent acquisition of resistance and/or virulence-associated plasmids by lineages that did not originally carry both trait sets. This is consistent with the plasmid-mediated mechanisms documented in the reviewed literature (e.g., IncFIIK2-NDM-1 conjugative plasmids driving nosocomial spread of hypervirulent carbapenem-resistant clones), and reinforces that plasmid surveillance is not a peripheral addition but a central mechanism explaining how this dangerous phenotype combination arises and could continue to spread across otherwise distinct lineages.

The consistently low profile of ST147 (zero virulence score, minimal resistance, zero plasmids) provides a useful negative control within the dataset, illustrating that not all K. pneumoniae lineages carry meaningful AMR or virulence burden — reinforcing that the convergence phenomenon is lineage-specific rather than a universal species trait.

6. Limitations

- Sample size (n=21) is modest relative to large-scale surveillance studies (some reviewed papers used 100-1,000+ isolates); findings should be interpreted as illustrative of a pattern rather than statistically definitive.
- Genomic sources span three countries but were not selected via random or systematic epidemiological sampling; some geographic and study-level bias is possible.
- This is an in silico (genomic) surveillance study; virulence and resistance scores reflect genetic potential as predicted by Kleborate's curated schemes, not laboratory-confirmed phenotypic testing.
- Recombination-aware phylogenetic refinement (e.g., Gubbins) was not performed; the tree reflects standard core-genome maximum-likelihood inference without explicit recombination filtering.
- Plasmid reconstruction from short-read/complete assemblies via MOB-suite provides strong evidence of plasmid presence and typing, but does not itself prove active horizontal transfer between isolates — this would require additional experimental or fine-scale comparative plasmid analysis.

7. Conclusion

This project applied an integrated genomic surveillance pipeline — spanning quality control, annotation, lineage typing, resistome and virulome profiling, plasmid reconstruction, pangenome analysis, and phylogenetics — to 21 publicly available human clinical K. pneumoniae genomes. The results demonstrate clear, quantifiable evidence of AMR-virulence convergence within specific lineages (ST2096 and ST11), occurring independently across distinct phylogenetic clades and associated with multi-plasmid carriage. These findings are consistent with an actively growing area of concern in the global AMR literature and illustrate how a well-scoped genomic surveillance workflow, applied even to a modest public dataset, can recover biologically meaningful and clinically relevant patterns.

8. References

Genomic literature reviewed (PMIDs):

- PMID 42070149 — Genomic analyses identify nosocomial transmission of ST23 carbapenem-resistant hypervirulent Klebsiella pneumoniae mediated by a conjugative IncFIIK2 NDM-1 plasmid.
- PMID 42590966 — Comparative genomics of carbapenem resistance and virulence associated with respiratory tract and bloodstream infections in carbapenem-resistant Klebsiella pneumoniae.
- PMID 42300588 — Genetic determinants of hypervirulence with attenuated pathogenicity in a blaNDM-1-encoding Klebsiella pneumoniae causing neonatal sepsis.
- PMID 42118687 — Clinical and genomic profiling of Klebsiella pneumoniae in liver abscesses.
- PMID 42359170 — Molecular characterisation of extended-Spectrum-Beta-lactamase-producing Klebsiella pneumoniae among children and chicken in the rural Korogwe District, Tanzania.
- PMID 41573971 — Are hospital environments an underestimated source for Gram-negative infections in critically ill patients? A non-multi-drug-resistant Klebsiella pneumoniae outbreak in an Irish intensive care unit.
- PMID 41343892 — Molecular epidemiological investigation of the carbapenemase-producing Enterobacterales isolates in Okayama prefecture, Japan.

- PMID 41451987 — Multidrug-resistant gram-negative bacteria in Spanish ICU patients: clinical and microbiological characterization (MURAN-UCI Project).

Tools cited:

- Seemann T (2014) Prokka: rapid prokaryotic genome annotation. Bioinformatics 30(14):2068-9.
- Lam MMC et al. Kleborate: a tool for genomic surveillance of Klebsiella pneumoniae and its related species complex.
- Robertson J, Nash JHE. MOB-suite: software tools for clustering, reconstruction and typing of plasmids from draft assemblies.
- Tonkin-Hill G et al. (2020) Producing polished prokaryotic pangenomes with the Panaroo pipeline. Genome Biology.
- Minh BQ et al. (2020) IQ-TREE 2: New models and efficient methods for phylogenetic inference in the genomic era. Molecular Biology and Evolution.
- Letunic I, Bork P (2024) Interactive Tree of Life (iTOL) v6. Nucleic Acids Research.

Appendix A: Full Genome Accession List

Accession	Strain	Source Study
GCA_055721705.1	KK22	Turkiye CRKP study
GCA_055721685.1	KK33	Turkiye CRKP study
GCA_055721665.1	KK35	Turkiye CRKP study
GCA_055721645.1	KK45	Turkiye CRKP study
GCA_055721625.1	KK47	Turkiye CRKP study
GCA_055721605.1	KK50	Turkiye CRKP study
GCA_055721585.1	KK53	Turkiye CRKP study
GCA_055721565.1	KK54	Turkiye CRKP study
GCA_055721545.1	KK55	Turkiye CRKP study
GCA_055721525.1	KK66	Turkiye CRKP study
GCA_053309215.1	KP1	Turkiye small study
GCA_053309195.1	KP2	Turkiye small study
GCA_053309175.1	KP3	Turkiye small study
GCA_053309155.1	KP4	Turkiye small study
GCA_054952645.1	SDS-K5	Pune, India
GCA_054952405.1	SDS-K9	Pune, India
GCA_054952445.1	SDS-K20	Pune, India
GCA_060056565.1	C8092	Peking, China
GCA_060056645.1	C2155	Peking, China
GCA_055485485.1	ST11-22-90	ST11 hospital study
GCA_055485515.1	ST11-16-116	ST11 hospital study

Appendix B: Tools and Software Versions

Phase	Tool	Version
1	Biopython	1.88
3	NCBI Datasets CLI	18.36.0
3	QUAST	5.2.0
3	Prokka	1.15.6
4	Kleborate	3.2.4
5	MOB-suite	3.1.9
6-7	Panaroo	1.8.0
7	MAFFT	(via Panaroo)
8	IQ-TREE	3.1.3
9	pandas	(Python integration script)
9	iTOL	web-based (v6)